# Section 5.5
# Complex Eigenvalues
Elementary Linear Algebra

(MATH 2250-03)

Spring, 2019

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Since the characteristic equation of an $n \times n$ matrix involves a polynomial of degree $n$, the equation always has exactly $n$ roots, counting multiplicities, provided that possibly complex roots are included. This section shows that if the characteristic equation of a real matrix A has some complex roots, then these roots provide critical information about A. The key is to let A act on the space $\mathbb{C}^n$ of $n$-tuples of complex numbers.

Our interest in $\mathbb{C}^n$ does not arise from a desire to "generalize" the results of the earlier chapters, although that would in fact open up significant new applications of linear algebra. Rather, this study of complex eigenvalues is essential in order to uncover "hidden" information about certain matrices with real entries that arise in a variety of real-life problems. Such problems include many real dynamical systems that involve periodic motion, vibration, or some type of rotation in space. The matrix eigenvalueeigenvector theory already developed for $\mathbb{R}^n$ applies equally well to $\mathbb{C}^n$. So a complex scalar $\lambda$ satisfies $\det\left(A - \lambda I\right) = 0$ if and only if there is a nonzero vector $x$ in $\mathbb{C}^n$ such that $Ax = \lambda x$. We call $\lambda$ a *(complex) eigenvalue* and $x$ a *(complex) eigenvector* corresponding to $\lambda$.

**Example 1.** If $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, then the linear transformation $x \mapsto Ax$ on $\mathbb{R}^2$ rotates the plane counterclockwise through a quarter-turn. The action of A is periodic, since after four quarter-turns, a vector is back where it started. Obviously, no nonzero vector is mapped into a multiple of itself, so A has no eigenvectors in $\mathbb{R}^2$ and hence no real eigenvalues. In fact, the characteristic equation of A is

$$\lambda^2 + 1 = 0.$$

The only roots are complex: $\lambda = i$ and $\lambda = -i$. However, if we permit A to act on $\mathbb{C}^2$, then

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} i \\ 1 \end{bmatrix} = i\begin{bmatrix} 1 \\ -i \end{bmatrix}$$
$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i \\ 1 \end{bmatrix} = -i\begin{bmatrix} 1 \\ i \end{bmatrix}$$

Thus $i$ and $-i$ are eigenvalues, with $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$ as corresponding eigenvectors.

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From the above example, it is clear that complex eigenvalues seem to have something to do with rotation. However, the example does not explain *why* the rotation occurs. The answer to this question rotation is hidden in the real and imaginary parts of a complex eigenvector.

## Real and Imaginary Parts of Vectors

The complex conjugate of a complex vector $x$ in $\mathbb{C}^n$ is the vector $\overline{x}$ in $\mathbb{C}^n$ whose entries are the complex conjugates of the entries in $x$. The real and imaginary parts of a complex vector $x$ are the vectors $\mathrm{Re}(x)$ and $\mathrm{Im}(x)$ in $\mathbb{R}^n$ formed from the real and imaginary parts of the entries of $x$.

**Example 2.** If $x = \begin{bmatrix} 3-i \\ i \\ 2+5i \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} + i \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$, then

$$\mathrm{Re}(x) = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}, \quad \mathrm{Im}(x) = \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}, \quad \text{and} \quad \overline{x} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} - i \begin{bmatrix} -1 \\ 1 \\ 6 \end{bmatrix} = \begin{bmatrix} 3+i \\ -i \\ 3-5i \end{bmatrix}.$$

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If B is an $m \times n$ matrix with possibly complex entries, then $\overline{\mathrm{B}}$ denotes the matrix whose entries are the complex conjugates of the entries in B. Properties of conjugates for complex numbers carry over to complex matrix algebra:

$$\overline{rx} = \overline{r}\ \overline{x}, \quad \overline{\mathrm{B}x} = \overline{\mathrm{B}}\ \overline{x}, \quad \overline{\mathrm{BC}} = \overline{\mathrm{B}}\ \overline{\mathrm{C}}, \quad \text{and } \overline{r\mathrm{B}} = \overline{r}\ \overline{\mathrm{B}}.$$

## Eigenvalues and Eigenvectors of a Real Matrix That Acts on $\mathbb{C}^n$

Let A be an $n \times n$ matrix whose entries are real. Then $\overline{\mathrm{A}x} = \overline{\mathrm{A}}\ \overline{x} = \mathrm{A}\overline{x}$. If $\lambda$ is an eigenvalue of A and $x$ is a corresponding eigenvector in $\mathbb{C}^n$, then

$$\mathrm{A}\overline{x} = \overline{\mathrm{A}x} = \overline{\lambda x} = \overline{\lambda}\ \overline{x}.$$

Hence $\overline{\lambda}$ is also an eigenvalue of A, with $\overline{x}$ a corresponding eigenvector. This shows that when A is real, *its complex eigenvalues occur in conjugate pairs*. (Here and elsewhere, we will use the term *complex eigenvalue* to refer to an eigenvalue $\lambda = a + bi$, with $b \neq 0$.)

**Example 3.** If $\mathrm{C} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where $a$ and $b$ are real numbers and not both zero, then the eigenvalues of C are $\lambda = a \pm bi$. Also, if $r = |\lambda| = \sqrt{a^2+b^2}$, then

$$\mathrm{C} = r \begin{bmatrix} a/r & -b/r \\ b/r & a/r \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{bmatrix}$$

where $\phi$ is the angle between the postive $x$-axis and line which passes through $(0,0)$ and $(a,b)$. Thus, the transformation $x \mapsto \mathrm{C}x$ may be viewed as the composition of a rotation through the angle $\phi$ and a scaling by $r$.

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The calculations in the above example can be carried out for any $2 \times 2$ real matrix A having a complex eigenvalue $\lambda$.

**Theorem 1.** *Let* A *be a real* $2 \times 2$ *matrix with a complex eigenvalue* $\lambda = a - bi$ *(with* $b \neq 0$*) and an associated eigenvector* $v \in \mathbb{C}^2$*. Then,*

$$\mathrm{A} = \mathrm{PCP}^{-1}, \quad \textit{where } \mathrm{P} = \begin{bmatrix} \mathit{Re}(v) & \mathit{Im}(v) \end{bmatrix} \quad \textit{and } \mathrm{C} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}.$$

The phenomenon displayed in the above example and theorem persists in higher dimensions as well. For brevity, however, we will not explore this further at the present moment.